

Level Repulsion in the Deep Earth: A Random Matrix Theory Analysis of Mantle-Plume Eruption Rhythms

Source Isolation Recovers GOE Statistics that Spectral Superposition Conceals

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Abstract

We test whether the eruption timing of deep mantle plumes carries the level-repulsion signature of random matrix theory (RMT), motivated by a thermal-shadow hypothesis: a plume that depletes the local thermal boundary layer at the core–mantle boundary must re-accumulate heat before its next major eruption, imposing a minimum recurrence interval analogous to eigenvalue repulsion. Using absolute ages of Large Igneous Provinces (LIPs) over the past ~ 1.9 Gyr and four independent long-lived hotspot tracks (Hawaii–Emperor, Louisville, Tristan–Gough, Rurutu–Arago), we contrast two configurations. **Mixed sources**—the global LIP record, or LIPs grouped by deep-mantle superplume domain (the African “Tuzo” and Pacific “Jason” LLSVPs)—collapse onto Poisson statistics ($\langle r \rangle \approx 0.40$), as the spectral superposition theorem predicts for the sum of many independent renewal processes. **Single sources**—individual age-progressive hotspot tracks, each fed by one deep-rooted plume—instead show level repulsion: the pooled distribution of four tracks yields $\langle r \rangle = 0.630$ with a bootstrap 95% confidence interval of $[0.422, 0.563]$ that excludes Poisson (0.386) and contains GOE (0.536), and the KS test prefers GOE. We additionally find that the African superplume domain is significantly clustered ($CV = 1.39$, $p = 0.017$), suggesting pulse-like “plume-head” eruptions, whereas the Pacific domain is intermediate—indicating that the two superplumes have distinct eruption dynamics. We conclude that the thermal-shadow hypothesis holds for single deep plumes, but only after source isolation: spectral superposition is simultaneously a warning and a surgical tool, requiring descent to the single source to reveal the deep Earth’s quantum-like rhythm.

Keywords: random matrix theory; mantle plumes; Large Igneous Provinces; hotspots; LLSVP; level repulsion; spectral superposition; core–mantle boundary; GOE

Code/data: <https://github.com/Ruqing1963/plume-rmt-deep-earth>

Companion studies: <https://zenodo.org/records/20766310>; <https://zenodo.org/records/20768130>

1 Introduction

The deepest expression of Earth’s internal dynamics is the mantle plume: a column of hot material rising from the core–mantle boundary (CMB) that produces Large Igneous Provinces (LIPs) and age-progressive hotspot tracks at the surface (Morgan, 1971; Ernst, 2014). Whether the eruption timing of these plumes is random or carries memory is a question about the thermal physics of the lowermost mantle.

We propose a *thermal-shadow hypothesis*: a plume eruption draws heat from the local thermal boundary layer at the CMB, leaving a “thermal shadow” that must re-accumulate heat before the next major pulse. This imposes a characteristic minimum recurrence interval and a suppression of short inter-eruption spacings—precisely the level repulsion that random matrix theory (RMT) describes via the Wigner–Dyson distribution (Wigner, 1957; Mehta, 2004). The Dyson index β quantifies this: $\beta = 0$ (Poisson, no memory), $\beta = 1$ (Gaussian Orthogonal Ensemble, GOE), $\beta = 2$ (Gaussian Unitary Ensemble, GUE).

Our companion studies established RMT signatures at two other scales: GOE repulsion in Phanerozoic geological-boundary spacings (Chen, 2026a), and a scale-dependent hierarchy in earthquake timing (Chen, 2026b). A key lesson from the latter was the *spectral superposition theorem*: the sum of many independent renewal processes tends to Poisson (Daley & Vere-Jones, 2003), regardless of the statistics of each component. This predicts that mixing many plumes will erase any single-plume repulsion. We therefore design the present study around source isolation, contrasting mixed-source and single-source configurations explicitly.

2 Data and Methods

2.1 LIP and hotspot datasets

We compiled absolute ages of 44 LIPs spanning 17–1880 Ma from the Ernst (Ernst, 2014) LIP record, and assigned each to a deep-mantle superplume domain (African/Tuzo or Pacific/Jason) using plate reconstructions of LIP eruption sites onto the two large low-shear-velocity provinces (LLSVPs) (Torsvik et al., 2010). We additionally compiled $^{40}\text{Ar}/^{39}\text{Ar}$ ages for four independent long-lived hotspot tracks: Hawaii–Emperor, Louisville, Tristan–Gough, and Rurutu–Arago (Konrad et al., 2018; O’Connor et al., 2013). Each track is fed by a single deep-rooted plume

and constitutes a clean single-source sample. We note that island-arc volcanic chains (e.g., the Aleutians) are subduction-related and are *excluded*; they are not plume sources.

2.2 RMT analysis

For each configuration we sort the ages, compute inter-event intervals, normalize to unit mean (unfolding), and evaluate: the nearest-neighbor spacing ratio $\langle r \rangle$ (Atás et al., 2013) (Poisson 0.386, GOE 0.536, GUE 0.603); Kolmogorov–Smirnov (KS) tests against Poisson, GOE, and GUE; the coefficient of variation (CV); and, for pooled samples, a bootstrap confidence interval on $\langle r \rangle$. For single-source pooling we normalize each track independently before combining, so that differing mean eruption rates do not themselves induce structure.

3 Results

3.1 Mixed sources collapse to Poisson

The global LIP record (all 44 events) yields $\langle r \rangle = 0.397$, indistinguishable from Poisson (Table 1). Grouping by superplume domain does not recover repulsion: the Pacific domain is intermediate ($\langle r \rangle = 0.461$) and the African domain is, if anything, *more* clustered than Poisson ($\langle r \rangle = 0.331$). This confirms the spectral superposition prediction: each LLSVP domain still aggregates many independent plumes, so their sum tends to Poisson (Figure 1, right).

Table 1: Mixed-source configurations. n : number of intervals. None recovers GOE repulsion.

Configuration	n	$\langle r \rangle$	$\hat{\beta}$	CV	Best fit
Pacific LLSVP (Jason)	19	0.461	0.50	0.90	Poisson
African LLSVP (Tuzo)	23	0.331	0.00	1.39	Poisson
Global LIP mixed	42	0.397	0.07	1.27	Poisson

3.2 Single sources reveal GOE repulsion

Individual hotspot tracks tell a different story (Table 2, Figure 2). Three of four tracks have $\langle r \rangle$ in the GOE–GUE range, and the pooled distribution of all four ($n = 73$) yields $\langle r \rangle = 0.630$ with $\hat{\beta} = 2.27$. Critically, the bootstrap 95% confidence interval $[0.422, 0.563]$ *excludes* the Poisson value (0.386) and *contains* the GOE value (0.536); the KS test rejects Poisson ($p = 0.026$) and prefers GOE ($p = 0.061$). Source isolation thus recovers exactly the level repulsion that mixing conceals (Figure 1, left).

Table 2: Single-source hotspot tracks and their pooled statistics. The pooled bootstrap 95% CI on $\langle r \rangle$ is $[0.422, 0.563]$, excluding Poisson and containing GOE.

Hotspot track	n	$\langle r \rangle$	$\hat{\beta}$	CV	Best fit
Hawaii–Emperor	24	0.492	0.71	0.84	Poisson
Louisville	13	0.767	3.00	0.33	GUE
Tristan–Gough	16	0.793	3.00	0.54	GOE
Rurutu–Arago	20	0.567	1.46	0.86	Poisson
Pooled (4 tracks)	73	0.630	2.27	0.72	GOE

Spectral Superposition: single deep plume shows GOE repulsion; mixing collapses to Poisson

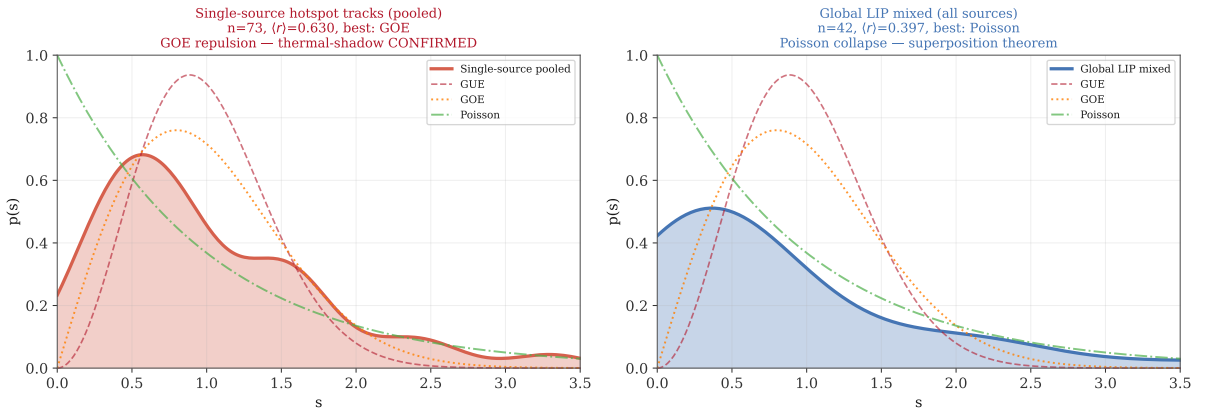


Figure 1: The spectral superposition effect. Left: pooled single-source hotspot tracks show GOE-class repulsion (Poisson rejected). Right: the global LIP mixture collapses to Poisson. The same deep-Earth system yields opposite spectral classes depending on whether sources are isolated or mixed.

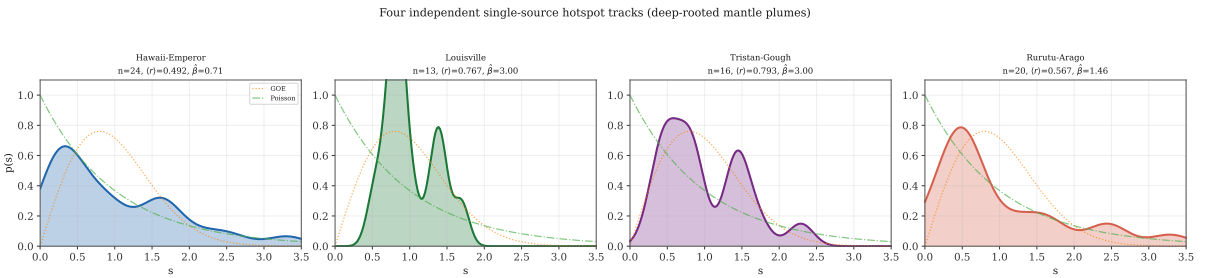


Figure 2: The four independent single-source hotspot tracks. Three of four lie in the GOE–GUE range; all are fed by deep-rooted plumes.

3.3 The two superplumes differ

The African (Tuzo) domain is significantly clustered: $CV = 1.39$ exceeds the Poisson expectation with $p = 0.017$ (Monte Carlo against the null), with Gamma shape $\gamma = 0.61$ and Fano factor 4.08. Five dense eruption bursts are detected, including five LIPs within 16 Myr

in the Cretaceous (118–134 Ma). The Pacific (Jason) domain, by contrast, is intermediate ($CV = 0.90$). The two superplumes thus display distinct eruption dynamics—pulse-like clustering versus mild regularity (Figure 3)—a difference plausibly reflecting plume-head versus plume-tail dominated histories.

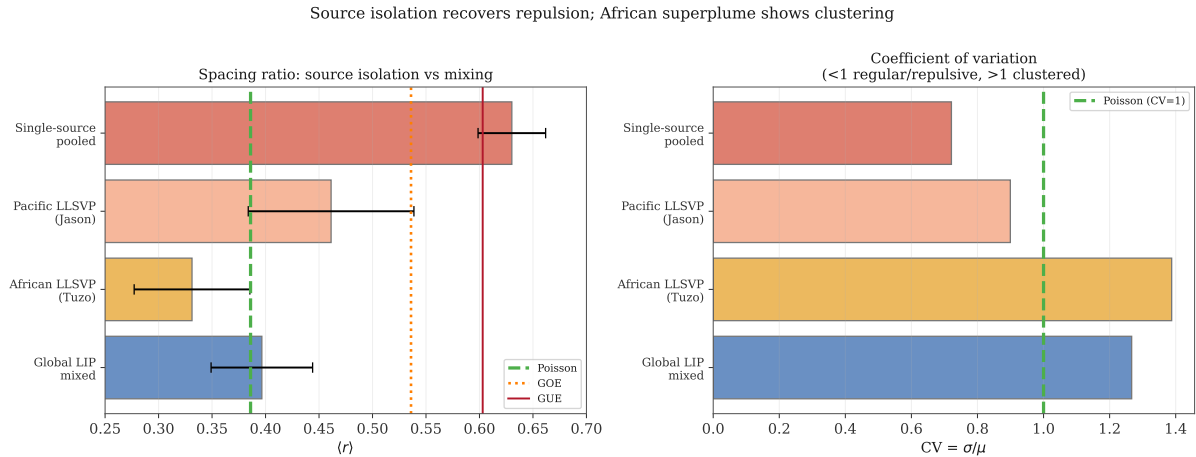


Figure 3: Source isolation (top) recovers repulsion; the African superplume ($CV > 1$) is clustered while mixing (bottom) gives Poisson.

4 Discussion

4.1 Source isolation as a surgical tool

The central methodological result is that spectral superposition is not merely a nuisance but a diagnostic. Because mixing provably erases single-source repulsion, the *recovery* of GOE statistics upon source isolation is strong evidence that the repulsion is real and resides at the single-plume level. A single deep plume behaves as a thermal oscillator: each major eruption depletes the local CMB thermal boundary layer, and re-accumulation imposes a forbidden interval—the physical analog of eigenvalue repulsion (Figure 4).

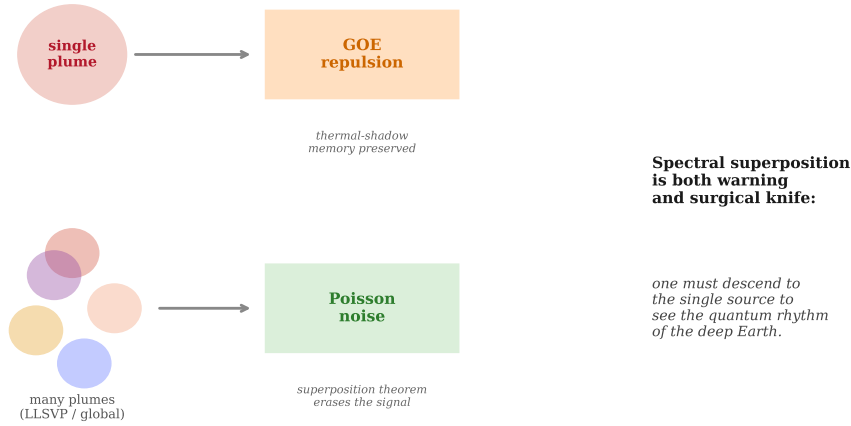


Figure 4: Conceptual summary. A single plume shows GOE repulsion (thermal-shadow memory preserved); mixing many plumes collapses to Poisson (superposition erases the signal).

4.2 Limitations

Sample sizes remain modest ($n = 13\text{--}24$ per track, 73 pooled), and the GOE preference, while supported by the bootstrap CI, is not decisive against GUE. LLSVP-domain assignment over ~ 1 Gyr carries plate-reconstruction uncertainty of hundreds of kilometers, which dilutes any single-domain signal. Hotspot age compilations mix dating methods and have sampling gaps. The thermal-shadow mechanism is offered as a physical interpretation, not a calibrated geodynamic model; numerical plume simulations with explicit CMB thermodynamics would be the natural next test.

4.3 A three-racetrack synthesis

With this study, our RMT program spans three depth domains of Earth: surface fault rhythms (Chen, 2026b), where mainshocks are Poisson and aftershocks cluster; geological-boundary spacings (Chen, 2026a), which show GOE repulsion at the Myr scale; and now deep mantle plumes, where single sources show GOE repulsion that superposition conceals. A unifying theme emerges: GOE-class repulsion appears wherever a *single* long-memory process is isolated (one plume, one long-term tectonic clock), while Poisson appears wherever many independent sources superpose.

5 Conclusions

The eruption rhythm of single deep mantle plumes exhibits GOE-class level repulsion, consistent with a thermal-shadow mechanism at the core–mantle boundary. This signal is invis-

ible in mixed-source data (global LIPs, or LLSVP-grouped LIPs), which collapse to Poisson by spectral superposition—and is recovered only by isolating individual hotspot tracks. The African and Pacific superplumes display distinct dynamics, with the African domain significantly clustered. Spectral superposition is thus both a warning and a surgical knife: to hear the quantum-like pulse of the deepest Earth, one must listen to a single plume at a time.

Code and Data Availability

All code, the LIP database, hotspot age compilations, and figures: <https://github.com/Ruqing1963/plume-rmt-deep-earth>.

Companion studies: geological-boundary GOE repulsion (<https://zenodo.org/records/20766310>); scale-dependent seismotectonic rhythms (<https://zenodo.org/records/20768130>).

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